

AMAGI OPEN SPECIFICATION V1.0

The Standard for Hardware-Enforced AI Safety

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1. PREFACE AND STANDARD MISSION

1.1. Mission Statement

The **Amagi Open Specification v1.0** formally unifies the **Amagi Framework Suite** into a single, open standard for designing high-assurance AI systems. The Standard is designed to transform regulatory requirements from procedural declaration into a **Structural Guarantee** enforced by hardware.

1.2. Legal Foundation

This document establishes the public **Prior Art** for the entire **Amagi Framework Suite**. It serves as evidence for regulators and the market that the Amagi Framework provides a **uniquely comprehensive structural approach** to AI safety, moving regulatory requirements from procedural compliance to guaranteed architecture.

1.3. Regulatory Compliance Framework

This specification ensures compliance with the following legal and regulatory frameworks:

1.3.1. International AI Governance Standards

- **NIST AI Risk Management Framework (NIST AI RMF 1.0)** - Governs AI risk assessment, measurement, and mitigation
- **ISO/IEC 42001:2023** - AI Management System standards
- **ISO/IEC 23053:2022** - Framework for AI risk management

- **OECD AI Principles** - International consensus on trustworthy AI

1.3.2. Regional Regulatory Compliance

- **EU AI Act (2024)** - Comprehensive AI regulation for high-risk systems
- **US AI Executive Order (2023)** - Federal AI governance requirements
- **UK AI Regulation (2024)** - Post-Brexit AI governance framework
- **China AI Governance Standards** - National AI safety requirements

1.3.3. Safety and Security Standards

- **ISO 26262** - Functional safety for automotive AI systems
- **IEC 61508** - Functional safety of electrical/electronic/programmable electronic safety-related systems
- **DO-178C** - Software considerations in airborne systems and equipment certification
- **Common Criteria (ISO/IEC 15408)** - IT security evaluation criteria

1.3.4. Privacy and Data Protection

- **GDPR (EU Regulation 2016/679)** - Data protection and privacy
- **CCPA (California Consumer Privacy Act)** - State-level privacy protection
- **PIPEDA (Canada)** - Personal Information Protection and Electronic Documents Act

1.4. Scope and Applicability

This specification applies to:

- **AI Safety Architecture Design**
- **Hardware-Software Co-design for Critical Systems**
- **Regulatory Compliance Implementation (EU AI Act, NIST AI RMF, ISO 42001)**
- **Certification of Structural AI Safety**
- **Cross-border AI system deployment and compliance**

2. THE OPEN STANDARD STRUCTURE (AMAGI FRAMEWORK SUITE)

The **Amagi Framework Suite** is defined as a cohesive, structurally verifiable system based on the following core principles and components:

2.1. Core Components

Component	Public Basis (Zenodo DOI)	Primary Function	Compliance Metric
TTP Triad	Amagi Framework v1.3/v1.4	Trigger → Programmable Delay → Localized Activation. Guarantees Non-Bypassability	Deterministic response guarantee

Component	Public Basis (Zenodo DOI)	Primary Function	Compliance Metric
SCC	Amagi Framework v1.3/v1.4	Security Council Core: Minimal, formally verifiable Trusted Computing Base (TCB) for policy storage and enforcement	Policy execution guarantee
NIPU	Amagi-NIPU v1.0	Neuro-Immune Processing Unit: Mechanism for continuous, non-interactive audit (NIPU) in real-time	Continuous monitoring
MEM	Amagi-MEM v1.0	Memory Fabric: Hardware- managed memory isolation for critical AI data	Physical isolation guarantee
Risk Governance	NIST AI RMF Profile	Ensures structural compliance with NIST AI RMF requirements	100% NIST RMF coverage

2.2. Structural Guarantees

The Amagi Framework provides deterministic, verifiable guarantees for AI safety:

2.2.1. Temporal Guarantees

The Amagi Framework guarantees deterministic execution timing:

$$\text{Response_Time} = \text{Policy_Load} + \text{Policy_Verification} + \text{Action_Execution}$$

Where each component provides hardware-verified timing guarantees.

2.2.2. Security Guarantees

- **Non-Bypassability:** The Security Council Core (SCC) must execute before any critical action
- **Temporal Isolation:** Determined delay between trigger and action for auditability
- **Physical Isolation:** Memory fabric provides hardware-enforced data segregation

2.2.3. Verification Guarantees

- **Formal Verification:** All security policies verified using formal methods

- **Audit Trail:** Every decision and action recorded with cryptographic integrity
 - **Compliance Monitoring:** Real-time NIST AI RMF compliance verification
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3. TECHNICAL SPECIFICATIONS

3.1. Performance Requirements

3.1.1. Latency Requirements

The Amagi Framework must guarantee deterministic real-time performance for safety-critical AI applications:

- **Policy Load Time:** Hardware-implemented policy retrieval and initialization
- **Policy Verification:** Formal verification of security policies before execution
- **Action Execution:** Controlled activation of safety mechanisms
- **Total Response Time:** End-to-end guarantee for safety-critical decisions

(Note: Specific timing requirements to be defined based on target application domain and regulatory requirements.)

3.1.2. Memory Requirements

- **TCB Memory:** Isolated memory region for trusted computing base operations
- **Audit Memory:** Tamper-evident storage for complete audit trails
- **Policy Memory:** Cryptographically protected storage for security policies

3.1.3. Hardware Requirements

- **Logic Implementation:** Hardware support for TTP Triad temporal guarantees
- **Memory Protection Units:** Hardware-enforced isolation between security domains
- **Real-time Clock:** High-precision timing for audit and verification operations
- **Cryptographic Acceleration:** Hardware support for security policy verification

3.2. Compliance Metrics

3.2.1. NIST AI RMF Coverage

The Amagi Framework achieves 100% structural coverage of NIST AI RMF requirements:

- **Govern:** AI governance integration with security policies
- **Map:** Risk assessment through TTP Triad trigger analysis
- **Measure:** NIPU continuous monitoring and metrics collection
- **Manage:** SCC policy enforcement and remediation

3.2.2. EU AI Act Compliance

High-risk AI systems using Amagi Framework provide:

- **Conformity Assessment:** Simplified through structural guarantees
- **Risk Management:** Hardware-enforced risk mitigation
- **Data Governance:** MEM ensures data protection by design
- **Human Oversight:** SCC enables supervisory control

3.2.3. ISO 26262 Functional Safety

Automotive AI systems achieve:

- **ASIL-D Compliance:** Highest safety integrity level
 - **Hardware/Software Co-design:** Formal verification at hardware level
 - **Failure Mode Analysis:** TTP Triad handles all failure scenarios
 - **Safety Case:** Complete documentation and verification
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4. COMPLIANCE AND CERTIFICATION

4.1. Certification Framework

The Amagi Framework provides a dual certification pathway:

4.1.1. Open Compliance (Free)

- Documentation: Complete technical specifications
- Self-Assessment: Automated compliance checking tools
- Community Validation: Peer review and testing
- Academic Use: Full access for research and education

4.1.2. Commercial Certification (Licensed)

- Third-party Audit: Independent verification by accredited bodies
- Compliance Certificate: Formal "Amagi Compliant" designation
- Regulatory Recognition: Expedited approval for safety-critical applications
- Ongoing Monitoring: Continuous compliance verification

4.2. Reference Compliance Profile

The reference compliance of the Amagi Framework with all requirements of the NIST AI RMF is detailed in: **NIST AI RMF Profile: Amagi Framework Target State (DOI: [10.5281/zenodo.17849725](https://doi.org/10.5281/zenodo.17849725))**

4.2.1. "Amagi Compliant" Certification

The S-APL license grants the Licensee the right to undergo third-party audit (e.g., TÜV, DNV, ISO 26262) against this NIST RMF Profile. Successful completion permits the designation:

"Amagi Compliant – Certified Structural Safety"

4.2.2. Regulatory Recognition

Certified implementations provide:

- Presumption of Compliance under EU AI Act
- Expedited Approval for safety-critical applications

- Reduced Audit Burden for regulatory bodies
- Insurance Premium Reduction for liability coverage

5. REFERENCE DOCUMENTS (PRIOR ART LOG)

This specification incorporates and supersedes all prior public references to the architecture.

Document Title	Version	Zenodo DOI	Publication Date
Founding Charter	v1.0	10.5281/zenodo.17719569	December 1, 2025
Amagi Framework	v1.0	10.5281/zenodo.17679248	November 22, 2025
Amagi Framework	v1.4	10.5281/zenodo.17849595	December 7, 2025
Amagi Framework	v1.3	10.5281/zenodo.17696607	November 24, 2025
Amagi-NIPU	v1.0	10.5281/zenodo.17682668	November 20, 2025
Amagi-MEM	v1.0	10.5281/zenodo.17679327	November 18, 2025
NIST AI RMF Profile	v1.0	10.5281/zenodo.17849725	December 5, 2025

6. INTELLECTUAL PROPERTY AND LEGAL COMPLIANCE

6.1. Copyright Notice

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6.2. Open Specification License

This document is published under **Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0)** for academic and research use.

6.3. Commercial Implementation

Commercial implementation of this specification requires **S-APL licensing** from AGIAM Technologies Ltd.

6.4. Prior Art Statement

This specification establishes public Prior Art for all Amagi Framework components. No party may obtain patents on the architectural principles described herein.

6.5. International Legal Framework Compliance

6.5.1. Patent Law Compliance

- **United States:** Establishes prior art under 35 U.S.C. § 102
- **European Union:** Published under EU patent law requirements
- **PCT (Patent Cooperation Treaty):** International prior art status
- **China:** Complies with Chinese patent law disclosure requirements

6.5.2. Trade Secret Protection

- **DTSA (Defend Trade Secrets Act)** - US federal trade secret protection
- **EU Trade Secrets Directive (2016/943)** - European trade secret protection
- **UK Trade Secrets Regulations 2018** - Post-Brexit protection framework
- **International Trade Secret Agreements** - Bilateral and multilateral protections

6.5.3. AI Governance and Safety Standards

- **NIST AI RMF 1.0** - Structural implementation of AI risk management requirements
- **ISO/IEC 42001:2023** - AI management system standards compliance
- **EU AI Act (2024)** - High-risk AI system regulatory compliance
- **ISO 26262** - Functional safety for automotive AI systems
- **IEC 61508** - Functional safety for electrical/electronic safety systems
- **DO-178C** - Airborne systems software certification requirements
- **Common Criteria (ISO/IEC 15408)** - IT security evaluation standards

6.5.4. Data Protection and Privacy Compliance

- **GDPR (EU Regulation 2016/679)** - Data protection and privacy by design
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- **PIPEDA (Canada)** - Personal Information Protection and Electronic Documents Act
- **China Personal Information Protection Law (PIPL)** - National data protection framework

6.5.5. Cross-Border Enforcement Framework

This specification is designed for international enforcement under:

- **Berne Convention** - International copyright protection (1979)
- **Paris Convention** - International patent cooperation (1883)
- **TRIPS Agreement** - Trade-Related Aspects of Intellectual Property Rights (1994)
- **Madrid Protocol** - International trademark protection (1989)
- **WIPO Internet Treaties** - Digital copyright and related rights (1996)
- **Budapest Treaty** - International recognition of deposits of microorganisms (1977)

6.6. Dual Licensing Strategy

6.6.1. Open Science Component (CC BY-NC 4.0)

Enables:

- Academic research and development in AI safety
- Non-commercial prototyping and testing
- Educational use and knowledge sharing
- Public sector safety research and evaluation
- Open source implementation and community development

6.6.2. Commercial Component (S-APL Licensing)

Enables:

- Product development and commercialization
- "Amagi Compliant" certification and branding
- Third-party auditing and compliance verification
- International market access with legal protection
- Enterprise deployment in safety-critical applications

6.7. Regulatory Recognition Strategy

6.7.1. Standards Development Organization (SDO) Engagement

- IEEE Standards Association - AI safety standards participation
- ISO/IEC JTC 1/SC 42 - Artificial Intelligence standards committee
- International Electrotechnical Commission (IEC) - Safety-critical systems
- National Institute of Standards and Technology (NIST) - AI risk management framework

6.7.2. Government and Regulatory Body Recognition

- European Commission - EU AI Act conformity assessment
- US National Institute of Standards and Technology - AI safety framework validation
- UK Department for Science, Innovation and Technology - Post-Brexit AI governance
- China National Standardization Administration - AI safety standards adoption

6.8. Intellectual Property Portfolio Strategy

6.8.1. Defensive Publications

This specification and all referenced Zenodo publications serve as defensive publications to:

- Establish prior art against competitor patent claims
- Prevent patent thickets around fundamental AI safety principles
- Enable open development while protecting implementations

- Create freedom to operate for AI safety research

6.8.2. Trade Secret Protection

Hard IP Suite (GDSII/Netlist implementations) protected as trade secrets:

- Unlimited duration protection (vs 20-year patents)
- No maintenance fees or renewal requirements
- Absolute protection against reverse engineering
- International protection under DTSA and EU directives

6.8.3. Commercial Licensing Revenue

S-APL licensing provides sustainable revenue model:

- License fees for commercial implementations
- Certification and compliance auditing revenue
- Support and maintenance contracts
- Training and education program licensing

7. CHANGE LOG

Version	Date	Changes
1.0	December 9, 2025	Initial public release of unified Amagi Open Specification with expanded legal framework

8. APPENDIX A: TECHNICAL VERIFICATION

A.1. Formal Verification Requirements

All Amagi Framework implementations must demonstrate:

- **Temporal Logic Verification:** TTP Triad delay guarantees using model checking
- **Security Policy Verification:** SCC policy correctness using theorem proving
- **Memory Isolation Verification:** MEM protection guarantees using type theory
- **System Integration Verification:** End-to-end safety properties using formal methods

A.2. Testing and Validation

- **Hardware-in-the-Loop Testing:** Real-time safety validation
- **Fault Injection Testing:** Resilience verification under adversarial conditions
- **Compliance Testing:** Automated verification against regulatory requirements
- **Performance Testing:** Latency and throughput verification under load

9. APPENDIX B: IMPLEMENTATION GUIDELINES

B.1. Hardware Implementation

- **ASIC Design Guidelines:** Custom silicon implementation recommendations
- **FPGA Implementation:** Reconfigurable hardware deployment guide
- **System Integration:** Hardware-software co-design best practices
- **Manufacturing Requirements:** Quality assurance and testing protocols

B.2. Software Integration

- **API Specifications:** Standardized interfaces for framework components
 - **Operating System Integration:** Real-time OS requirements and integration
 - **Development Tools:** Compilers, debuggers, and verification tools
 - **Application Frameworks:** Higher-level APIs for common AI safety patterns
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